Stability of a prescribed repeatable completion of a quantum measurement

Christopher Gardner

1 November 2026

External-review draft. The argument below gives an explicit quantitative specialization of established constructions and inequalities. The supplied reduction, its usefulness, and publication priority remain subjects for independent review.

Abstract

Let $E = (E_j)$ be a finite positive-operator-valued measurement (POVM), and let P_j be its full spectral sectors at or above $1 - \delta$, where $0 < \delta < 1/2$. Assume that each nonzero effect has a nonzero sector and that the sector synthesis operator is injective, with least singular value α. Its unweighted polar isometry determines orthogonal sectors Q_j. We prove that the prescribed completion $F_j = Q_j + RE_jR$, where $R = I - \sum_j Q_j$, admits a repeatable instrument and satisfies

$$\|\mathcal{M}_E - \mathcal{M}_F\|_\diamond \leq 2\min\{1, \sqrt{1-\alpha^2} + \sqrt{\delta}\}.$$

The proof reduces to polar geometry, a classical Jordan-product inequality, and pure-state trace distance after an explicit labelled-dilation calculation. For orthogonal input sectors, we determine the exact worst-case envelope $2\sqrt{\delta}/(2 - \sqrt{\delta})$ and exhibit an attaining family. For general packing, a residual decomposition and a coupled simplex lower construction determine the sharp joint leading envelope $\sqrt{1-\alpha^2+\delta}$ as $(\alpha, \delta) \to (1, 0)$, uniformly over dimension and outcome count. Exact rational certificates show both a strict reference advantage and the failure of a natural finite-error joint extension. A corollary corrects separated output records to an exactly idempotent channel. Both sector constructions are established least-squares measurements; no new general operator lemma or literature priority is claimed. The exact joint finite-error envelope remains open.

1. Question and scope

A finite POVM can admit a repeatable instrument without being a projection-valued measurement (PVM). The relevant criterion is that every nonzero effect have eigenvalue one (Busch et al. 1995). Such measurements have orthogonal unit eigenspaces and may retain an arbitrary POVM on their common orthogonal complement. We study a particular way to select a nearby measurement in this class while retaining the input space, labels, and compressed residual effects.

The construction uses the polar factor of a synthesis operator, a standard orthogonalization device (Higham 1986). Its sector vectors are exactly the unweighted least-squares measurement of Eldar–Forney; the weighted counterpart below is their weighted version (Eldar and Forney 2001). The issue addressed here is quantitative control of a prescribed residual extension in the full diamond norm. The hypothesis concerns near-unit sectors and their simultaneous linear independence; it is not a global almost-projectivity assumption.

The distinction from earlier work is limited. De la Salle's orthogonalization proof also contains a closely related weighted polar construction (Salle 2022). Dilation continuity is standard (Kretschmann et al. 2008), and the noncommuting matrix inequality used below is a special case of the Strang–de Bruijn Jordan-product bounds (Bruijn 1980). No additional general operator lemma is needed: the measurement-specific work is the labelled dilation and residual compression, followed by parameter substitution. Section 5 makes the reduction explicit. This is a quantitative specialization with a complete argument.

2. Setting and main statement

Let $\mathcal{H} \neq \{0\}$ be a finite-dimensional complex Hilbert space. Write $\|\cdot\|$ for operator norm and $\|\cdot\|_1$ for trace norm. A POVM $E = (E_j)_{j=1}^m$, $m \geq 1$, consists of positive operators summing to I. Its measurement channel is

$$\mathcal{M}_E(\rho) = \sum_j \mathrm{Tr}(E_j\rho)|j\rangle\langle j|.$$

We use the unhalved diamond norm; all state distances below likewise use the unhalved trace norm. For a difference of channels the diamond norm equals the supremum of output trace-norm difference over density operators with an arbitrary reference system; reference-assisted pure inputs suffice (Watrous 2018, Theorem 3.51). Consequently its maximum is two. Channels are completely positive and trace preserving (CPTP); their adjoints are unital completely positive (UCP).

Fix $0 < \delta < 1/2$ and set

$$P_j = 1_{[1-\delta,1]}(E_j), \qquad \mathcal{K}_j = \operatorname{ran} P_j, \qquad \mathcal{K} = \bigoplus_j \mathcal{K}_j. \tag{1}$$

Assume $P_j \neq 0$ whenever $E_j \neq 0$. A zero effect has $P_j = 0$ and may retain its label. Define

$$T : \mathcal{K} \longrightarrow \mathcal{H}, \qquad T(x_1, \ldots, x_m) = \sum_j x_j, \qquad \alpha = \sigma_{\min}(T) > 0. \tag{2}$$

Thus T is injective. Write $T = UH$, with $H = |T|$, and let C_j be the coordinate projection of $\mathcal{K}$ onto $\mathcal{K}_j$. Put

$$Q_j = UC_jU^*, \qquad S = UU^*, \qquad R = I - S, \qquad F_j = Q_j + RE_jR. \tag{3}$$

Theorem 1 (prescribed completion). Under (1)–(2), the Q_j are mutually orthogonal, have the ranks of P_j, and obey

$$\|P_j - Q_j\| \leq a, \qquad a := \sqrt{1-\alpha^2}. \tag{4}$$

The family F in (3) is a POVM with full unit eigenspaces Q_j, and

$$\begin{gathered} F_iQ_j = \delta_{ij}Q_j, \\ \gamma_F := \min_j \left[1 - \|(I-Q_j)F_j(I-Q_j)\|\right] \geq \delta. \end{gathered} \tag{5}$$

It admits a repeatable instrument and satisfies

$$\boxed{\|\mathcal{M}_E - \mathcal{M}_F\|_\diamond \leq 2\min\{1, a + \sqrt{\delta}\}.} \tag{6}$$

More precisely, writing $d_0 = \sqrt{1-\delta}$ and $M = \|H\|$, the sufficient condition

$$(M - \alpha)(1 - d_0) \leq 4\alpha d_0 \tag{7}$$

implies

$$\|\mathcal{M}_E - \mathcal{M}_F\|_\diamond \le 2\sqrt{1-\alpha^2(1-\delta)}. \tag{8}$$

The constants contain no explicit dimension, outcome-count, or sector-rank factor. The domain remains restrictive: $\sum_j \operatorname{rank} P_j \le \dim \mathcal{H}$, so there are at most $\dim \mathcal{H}$ nonzero outcomes. The estimate does not apply to generic overcomplete POVMs.

3. Proof of Theorem 1

3.1. Geometry and the residual gap

The polar factor U is an isometry with $\operatorname{ran} U = \operatorname{ran} T = \sum_j \operatorname{ran} P_j$. Thus $P_j R = 0$, and the Q_j are orthogonal and rank preserving. A unit vector supported in one nonzero coordinate sector has $\|Tx\| = 1$, proving $\alpha \le 1$. The spectral inequality $E_j \ge (1-\delta)P_j$ gives

$$(1-\delta)TT^* = (1-\delta)\sum_j P_j \le I, \qquad M \le d_0^{-1}. \tag{9}$$

For a nonzero sector let $\iota_j : \mathcal{K}_j \to \mathcal{K}$ be the coordinate inclusion, $T_j = T\iota_j$, and $U_j = U\iota_j$. Both T_j and U_j are isometries, and

$$T_j^* U_j = \iota_j^* H \iota_j \ge \alpha I_{\mathcal{K}_j}.$$

Hence their overlap has smallest singular value at least α. It follows that $\|(I-P_j)Q_j\|$ and $\|(I-Q_j)P_j\|$ are at most a. For two orthogonal projections their difference norm is the maximum of these two quantities, proving (4). One may verify this identity by squaring the difference and taking its two diagonal blocks relative to the first projection. Zero sectors are immediate.

Each F_j is positive, and

$$\sum_j F_j = S + R\Big(\sum_j E_j\Big)R = I.$$

Equation (5)'s sector action follows from orthogonality. Fullness of the spectral sector gives

$$E_j \le P_j + (1-\delta)(I-P_j), \qquad RE_jR \le (1-\delta)R. \tag{10}$$

On S, the effect F_j is Q_j; on R, its norm is at most $1-\delta$. Thus Q_j is its full unit eigenspace and (5) holds, including zero effects. Choosing a density operator τ_j supported in each nonzero Q_j gives the instrument $\mathcal{I}_j(\rho) = \operatorname{Tr}(F_j\rho)\tau_j$. For a zero effect set $\mathcal{I}_j = 0$. Since $\operatorname{Tr}(F_i\tau_j) = \delta_{ij}$, a repetition returns the same outcome.

3.2. An explicit labelled dilation

On $\mathcal{Y} = \bigoplus_j \mathcal{H}$, let Λ_j project onto label j, and define

$$\begin{aligned} W &: \mathcal{H} \to \mathcal{Y}, \quad Wz = (\sqrt{E_j}z)_j, \\ J &: \mathcal{K} \to \mathcal{Y}, \quad J(x_j)_j = (x_j)_j. \end{aligned} \tag{11}$$

In particular, J embeds the original sector vectors. Both W and J are isometries. Spectral commutation allows the block operator

$$D = \bigoplus_j \sqrt{E_j}\big|_{\mathcal{K}_j}, \qquad d_0 I \le D \le I.$$

Directly from their domains,

$$W^*J = TD = UHD, \qquad J^*\Lambda_j W R = 0 \quad \text{for every } j. \tag{12}$$

Indeed, the only possible nonzero coordinate of the second expression is $P_j\sqrt{E_j}R = \sqrt{E_j}P_jR = 0$.

Set

$$V = JU^* + WR. \tag{13}$$

The cross terms in V^*V vanish, so $V^*V = S + R = I$. The labelwise identities in (12), together with $J^*\Lambda_j J = C_j$, give

$$V^*\Lambda_j V = UC_jU^* + RE_jR = F_j, \qquad W^*V = UHDU^* + R. \tag{14}$$

These identities concern the particular completion (3).

3.3. The overlap estimate

Lemma 2 (classical Jordan-product specialization). If $\alpha I \le H \le MI$, $d_0 I \le D \le I$, and (7) holds, then $\mathrm{Re}(HD) \ge \alpha d_0 I$.

Proof. Set $X = H - \alpha I$, $Y = D - d_0 I$, $h = M - \alpha$, and $r = 1 - d_0$. For $s > 0$, positivity of $(\sqrt{s}X + s^{-1/2}Y)^2$ implies

$$XY + YX \ge -sX^2 - s^{-1}Y^2 \ge -shX - (r/s)Y.$$

Therefore

$$\mathrm{Re}(HD) - \alpha d_0 I \ge (d_0 - sh/2)X + (\alpha - r/(2s))Y.$$

For $h > 0$, take $s = 2d_0/h$; both coefficients are nonnegative by (7). For $h = 0$, $H = \alpha I$ and the assertion is immediate. □

For attribution, put $u = (M - \alpha)/(M + \alpha)$ and $v = (1 - d_0)/(1 + d_0)$. Condition (7) is exactly $u + v \le 1$; the corresponding branch of de Bruijn's Theorem 3.2 and Sections 4–5 gives the lower bound $(M + \alpha)(1 + d_0)(1 - u)(1 - v)/4 = \alpha d_0$ (Bruijn 1980). The proof above is included for completeness.

By (14) and $\alpha d_0 \le 1$, Lemma 2 yields

$$\mathrm{Re}(W^*V) \ge \alpha d_0 S + R \ge \alpha d_0 I.$$

For any normalized $\psi \in \mathcal{H} \otimes \mathcal{A}$, the dilated vectors $(W \otimes I_{\mathcal{A}})\psi$ and $(V \otimes I_{\mathcal{A}})\psi$ have overlap of modulus at least αd_0. The trace distance of their pure density operators is consequently at most $2\sqrt{1 - \alpha^2 d_0^2}$. Apply to both the same CPTP label readout $\mathcal{L}(X) = \sum_j \mathrm{Tr}(\Lambda_j X)|j\rangle\langle j|$, with the identity on $\mathcal{A}$. Trace-norm contractivity and the reference-assisted characterization prove (8).

To obtain (6), write $b = \sqrt{\delta}$. If $a + b \ge 1$, use the universal bound two. Otherwise $a < 1 - b$, so $\alpha^2 > 2b - b^2 > b^2$, hence $\alpha > b$. By (9),

$$(M - \alpha)(1 - d_0) \le M(1 - d_0) \le \frac{\delta}{d_0(1 + d_0)} \le \delta < 4\alpha d_0.$$

Here $d_0 > 1/\sqrt{2}$, and $b < 4d_0$. Thus (7) holds, and

$$2\sqrt{1 - \alpha^2(1 - \delta)} = 2\sqrt{a^2 + \alpha^2\delta} \le 2(a + b).$$

This completes the proof of Theorem 1. □

4. Correction of supplied records

Let σ_j be density operators on the same $\mathcal{H}$, and set

$$\Phi(\rho) = \sum_j \operatorname{Tr}(E_j\rho)\sigma_j, \qquad \eta = \|\Phi^2 - \Phi\|_\diamond.$$

For this section discard zero effects and their unused records. Assume at least two remaining labels and define

$$\begin{aligned} \kappa &= \min_j \frac{1}{2} \inf_{\omega \in \operatorname{conv}\{\sigma_i : i \neq j\}} \|\sigma_j - \omega\|_1 > 0, \\ e &= \max_j (1 - \lambda_{\max}(E_j)). \end{aligned} \tag{15}$$

The main hypotheses give $0 \le e \le \delta < 1/2$. With one remaining label the channel is constant and already idempotent.

Corollary 3 (record correction). Under Theorem 1 and (15), there are explicit states τ_j supported in Q_j such that $\Pi(\rho) = \sum_j \operatorname{Tr}(F_j\rho)\tau_j$ is exactly idempotent. For any valid measurement movement bound B_{move},

$$\boxed{\|\Phi - \Pi\|_\diamond \le \min\left\{2,\ B_{\text{move}} + 2a + \sqrt{\frac{2\eta}{\kappa\delta(1-2e)}}\right\}.} \tag{16}$$

In particular,

$$\|\Phi - \Pi\|_\diamond \le \min\left\{2,\ 4a + 2\sqrt{\delta} + \sqrt{\frac{2\eta}{\kappa\delta(1-2\delta)}}\right\}. \tag{17}$$

Proof. Put $v_j = \Phi(\sigma_j) - \sigma_j$ and $D_{\text{rec}} = \max_j \|v_j\|_1$. The action of $\Phi^2 - \Phi$ on a reference-assisted state is $\sum_j v_j \otimes B_j$, where the conditional operators B_j are positive and $\sum_j \operatorname{Tr} B_j = 1$. Thus $\eta \le D_{\text{rec}}$. Choose a maximizing label and an input whose probability of that label is at least $1 - e$. The reverse triangle inequality gives

$$(1 - 2e)D_{\text{rec}} \le \eta \le D_{\text{rec}}. \tag{18}$$

Writing $a_j = \operatorname{Tr}(E_j\sigma_j)$, when $a_j < 1$ the other outcomes yield a state ω_j in the competing convex hull with $v_j = (1 - a_j)(\omega_j - \sigma_j)$. The case $a_j = 1$ is immediate. Hence

$$2\kappa(1 - a_j) \le \|v_j\|_1, \qquad 1 - \operatorname{Tr}(P_j\sigma_j) \le q := \frac{\eta}{2\kappa\delta(1-2e)}, \tag{19}$$

where the second inequality uses (10).

For a purification of σ_j, the triangle inequality and (4) imply

$$\sqrt{1 - \operatorname{Tr}(Q_j\sigma_j)} \le \sqrt{1 - \operatorname{Tr}(P_j\sigma_j)} + \|P_j - Q_j\| \le \sqrt{q} + a.$$

If $z_j = \operatorname{Tr}(Q_j\sigma_j) > 0$, set $\tau_j = Q_j\sigma_jQ_j/z_j$. Projecting the purification gives a normalized vector with overlap $\sqrt{z_j}$ with the original; the pure-state trace distance formula and partial trace therefore give

$$\|\sigma_j - \tau_j\|_1 \le 2\sqrt{1 - z_j} \le 2\sqrt{q} + 2a. \tag{20}$$

This is the standard normalized gentle-projection estimate (Watrous 2018, Corollary 3.15 and Theorem 3.33). If $z_j = 0$, the preceding inequality forces $\sqrt{q} + a \ge 1$; choose $\tau_j = Q_j/\operatorname{rank} Q_j$, which satisfies (20).

Now $\mathrm{Tr}(F_i\tau_j) = \delta_{ij}$, so $\Pi^2 = \Pi$. Keeping the records σ_j while changing E to F costs at most B_{move}, by common CPTP post-processing. Keeping F while changing its records costs at most $\max_j \|\sigma_j - \tau_j\|_1$, also for inputs with a reference. Combining these facts proves (16). Theorem 1 and $e \le \delta$ give (17). □

For a concrete nontrivial regime, take $0 < \varepsilon < 1/16$ and $E_0 = \mathrm{diag}(1-\varepsilon, \varepsilon)$, $E_1 = I - E_0$, and the two computational-basis records. Then $\alpha = \kappa = 1$, $e = \varepsilon$, $\eta = 2\varepsilon(1-2\varepsilon)$, and the constructed channel is complete dephasing, at distance 2ε. Choosing $\delta = \sqrt{\varepsilon}$ in (16) with $B_{\mathrm{move}} = 2\sqrt{\delta}$ gives $4\varepsilon^{1/4}$, which is below two for $\varepsilon < 1/16$. The certificate is useful but can be very conservative.

5. Relationship to existing constructions

Choose an eigenbasis within every full sector and write the concatenated unit vectors as the columns of T. In these coordinates $D = \mathrm{diag}(\sqrt{\lambda_a})$, with λ_a the selected effect eigenvalues. Eldar–Forney's unweighted and weighted least-squares formulas are exactly

$$U = T(T^*T)^{-1/2}, \qquad \widehat{U} = TD(DT^*TD)^{-1/2}$$

(Eldar and Forney 2001, Theorem 2 and Section V). The weighted formula is also the square-root measurement for priors proportional to λ_a. Grouping columns by sector recovers Q_j and $\widehat{Q}_j$, including higher-rank sectors. These formulas optimize vector least-squares objectives; they do not assert optimal distance from the original measurement channel. The extension RE_jR and its channel comparison must still be checked.

Proposition 4 (weighted comparison). Keep the hypotheses and notation of Theorem 1, but let $TD = \widehat{U}|TD|$. Then, without condition (7),

$$\begin{aligned} \widehat{F}_j &= \widehat{U} C_j \widehat{U}^* + RE_jR, \\ \|\mathcal{M}_E - \mathcal{M}_{\widehat{F}}\|_\diamond &\le 2\sqrt{1 - \alpha^2(1-\delta)}. \end{aligned} \tag{21}$$

Proof. The invertibility of D gives $\mathrm{ran}(TD) = \mathrm{ran}\, T$ and $|TD| \ge \alpha d_0 I$. Thus $\widehat{U}$ has the same range projection S. Equations (12)–(14), with $\widehat{V} = J\widehat{U}^* + WR$, give the isometry and effect identities, while

$$W^*\widehat{V} = \widehat{U}|TD|\widehat{U}^* + R \ge \alpha d_0 I.$$

The same reference-assisted overlap argument proves (21). □

This is a reconstructed application of established polar alignment, not a claim that a cited paper states (21) verbatim. Its connection to de la Salle's proof is explicit. On $S\mathcal{H}$, put $A_j = SE_jS$. The P_j commute with their A_j, have total rank $\dim S\mathcal{H}$, and satisfy

$$\sum_j P_j A_j \ge (1-\delta)\alpha^2 I_{S\mathcal{H}}.$$

They can therefore replace the state-selected projections in the finite proof's polar step. That column becomes $x = (P_j\sqrt{A_j})_j = DT^*$; its adjoint is TD (Salle 2022, Lemma 3.1 and Section 3). The residual extension and overlap calculation above make the application uniform. Broad existence of a nearby same-label repeatable measurement consequently has a short derivation from this machinery. The weighted construction generally differs from (3); (21) does not prove Theorem 1's estimates for that prescribed map.

KSW's continuity theorem supplies the generic dilation-to-channel comparison (Kretschmann et al. 2008, Proposition 3). In tracking the sharper constant in (8), our proof uses pure-state trace distance directly: overlap at least c gives $2\sqrt{1-c^2}$; applying only $2\|W - V\|$ would give $2\sqrt{2(1-c)}$. Both are established estimates. Kotowski–Oszmaniec's dimension-deficient

Naimark construction proves projective simulability for a different complement-noise completion, without requiring unit eigenspaces or asserting the present diamond estimate (Kotowski and Oszmaniec 2025, Theorem 6).

The exact record endpoint also has established structure. On the algebra of active labels define $\Delta(f) = \sum_j f_j F_j$ and $\Upsilon(X) = (\mathrm{Tr}(\tau_j X))_j$. They are UCP maps with $\Upsilon\Delta = I$ and $\Pi^* = \Delta\Upsilon$, a commutative instance of exact encoding/decoding structure (Kitaev 2025, Theorem 1.2). Kitaev's Theorem 12.3 gives general approximate factors; it does not directly supply this particular exact correction of the original effects and records. The additional specification here is quantitative and applies to a substantially narrower class.

The reduction of the unweighted estimate is complete: (12)–(14) identify its dilation, the classical matrix bound supplies the overlap, and the pure-state formula and scalar case split prove (6). The sector estimate is principal-angle geometry. The record proof adds the reverse triangle inequality, convex-hull separation, and normalized gentle projection. Thus the bounds require no additional general lemma. The remaining questions are whether this explicit specialization is already documented and whether its quantitative control and limitations are useful. The certified comparison below distinguishes the two targets, but does not establish originality.

6. Counterexamples and limitations

6.1. Repeatable approximation differs from projective approximation

On $\mathbb{C}^3$, let

$$E_0(t) = \mathrm{diag}(1-t, t, 1/2), \quad E_1(t) = I - E_0(t), \qquad 0 < t < 1/2.$$

For $\delta = t$, the sectors are the first two coordinate lines and $\alpha = 1$. The prescribed completion is $F_0 = \mathrm{diag}(1, 0, 1/2)$, $F_1 = I - F_0$, with distance $2t$. However, its input measurement has distance exactly one to the nearest binary PVM. For any projection L, ordered-eigenvalue perturbation gives $\|E_0(t) - L\| \geq 1/2$, because the middle eigenvalue of $E_0(t)$ is $1/2$ whereas that of L is zero or one. Binary measurement-channel distance is $2\|E_0(t) - L\|$; the upper bound one is attained by $L = \mathrm{diag}(1, 0, 0)$. The binary norm identity follows from the norm of the Hermitian functional $\rho \mapsto \mathrm{Tr}((E_0 - L)\rho)$; an extremal eigenvector attains it, and reference systems do not increase it. Also $\|I - \sum_j E_j(t)^2\| = 1/2$.

Thus small δ and perfect packing do not imply global near-projectivity. The example does not preclude the packed-subspace reduction in Section 5.

6.2. A square-root threshold obstruction

Proposition 5. For every $0 < \delta < 1/10$, there is a three-outcome POVM on $\mathbb{C}^4$ satisfying Theorem 1 with $\alpha = 1$, such that every repeatable POVM G on the same space and with the same three labels obeys

$$\|\mathcal{M}_E - \mathcal{M}_G\|_\diamond \geq \frac{\sqrt{\delta}}{4\sqrt{3}}. \tag{22}$$

Proof. In the basis $e_1, \ldots, e_4$, put $t = \sqrt{\delta}/4$ and

$$E_1 = \begin{pmatrix} 1-\delta/2 & 0 & 0 & 0 \\ 0 & \delta/2 & 0 & t \\ 0 & 0 & 0 & 0 \\ 0 & t & 0 & 1/4 \end{pmatrix}, \tag{23}$$

$$E_2 = \mathrm{diag}(\delta/2, 1-\delta, \delta/2, 1/2), \quad E_3 = I - E_1 - E_2.$$

The nontrivial low 2×2 blocks of E_1, E_3 have determinant $\delta/16 > 0$ and trace $1/4+\delta/2 < 3/10$. The remaining entries are nonnegative. Thus these are POVM effects, and their full near-unit sectors are respectively $\mathbb{C}e_1, \mathbb{C}e_2, \mathbb{C}e_3$; hence $\alpha = 1$.

Set $\rho_0 = \max_j \|E_j - G_j\|$. If $\rho_0 \le 1/4$, then $G_2 \neq 0$, since $\|E_2\| = 1-\delta > 9/10$. Repeatability supplies a unit vector v with $G_2 v = v$, and positivity of the other effects gives $G_1 v = 0$. Write $v = a_2 e_2 + b_2$, $b_2 \perp e_2$. Since $I - E_2 \ge I/2$ on $e_2^{\perp}$,

$$\|(I-E_2)v\| \le \rho_0 \quad \Longrightarrow \quad \|b_2\| \le 2\rho_0, \quad |a_2| \ge \sqrt{3}/2.$$

Moreover, $\|E_1 v\| \le \rho_0$, $\|E_1\| \le 1$, and $\|E_1 e_2\| \ge t$. Therefore

$$|a_2| t \le \|a_2 E_1 e_2\| \le \rho_0 + \|b_2\| \le 3\rho_0,$$

so $\rho_0 \ge t/(2\sqrt{3})$. For any two POVMs, a state maximizing one effect difference gives $\|\mathcal{M}_E - \mathcal{M}_G\|_\diamond \ge 2\rho_0$: the other probability differences sum to its negative. This proves (22). If $\rho_0 > 1/4$, the same conclusion follows from distance greater than $1/2$. □

The proposition establishes the necessity of square-root threshold order for this same-space, same-label target class. It does not establish optimality of the constant two, the dependence on α, or the joint bound.

6.3. Neither completion uniformly dominates the other

Proposition 6 (computer-assisted comparison). On $\mathbb{C}^4$, set

$$E_1(t) = \frac{1}{40}\begin{pmatrix} 32 & 0 & 3 & 1 \\ 0 & 2 & 2 & 2 \\ 3 & 2 & 4 & 4 \\ 1 & 2 & 4 & 8 \end{pmatrix}, \qquad E_2(t) = \frac{1}{40}\begin{pmatrix} 4 & 2 & -2 & t \\ 2 & 34 & -3 & -1 \\ -2 & -3 & 2 & -2 \\ t & -1 & -2 & 12 \end{pmatrix}, \tag{24}$$
$$E_3(t) = I - E_1(t) - E_2(t), \qquad 2 \le t \le 4, \qquad \delta = 1/4.$$

This family satisfies Theorem 1 throughout the interval, with sector ranks $(1,1,1)$ and $\alpha(t) > 9/10$. Let $\nu_u(t)$ and $\nu_w(t)$ be the full diamond distances from the original measurement to the unweighted and weighted completions, and put $g(t) = \nu_w(t) - \nu_u(t)$. Exact certificates give

$$g(2) > 5\times 10^{-4}, \qquad g(4) < -9\times 10^{-5}. \tag{25}$$

The distances are continuous, and at least one equality point satisfies

$$2.33690021 < t_* < 2.33690023, \qquad g(t_*) = 0. \tag{26}$$

Uniqueness and absence of other equality points are not asserted.

Certificate argument. The accompanying rational data verify positivity at $t = 2, 4$, hence throughout the affine family. Designated diagonal entries give lower bounds $4/5, 17/20, 17/20$ for the top eigenvalues; the sums of the remaining eigenvalues are at most $7/20, 9/20, 7/10$, respectively. These bounds isolate exactly one eigenvalue above $3/4$ in each effect. The uniform spectral-residual and polar certificate proves $\alpha(t) > 9/10$ on the entire interval, not merely on a finite grid.

For each certified parameter and completion, the data specify rational effects G_j, a rigorous bound $\epsilon \ge \sum_j \|F_j - G_j\|$, and exact primal/dual feasible SDP witnesses with values L, U for $\mathcal{M}_E - \mathcal{M}_G$. Thus $L - \epsilon \le \nu \le U + \epsilon$, including reference systems. These enclosures prove (25) and strict opposite signs at the two endpoints in (26). Uniform spectral isolation and synthesis injectivity imply continuity, so the intermediate value theorem proves (26). The certificate

supplement links the data, the analytic error bounds and uniform argument, and the exact standard-library verifier. It is part of the computer-assisted proof of this proposition. □

The additive theorem bound stays below $1 + \sqrt{19}/5 < 2$ along this path. A stronger weighted upper envelope therefore does not imply a smaller actual weighted distance. These comparisons do not prove either completion optimal among all repeatable targets, sharpness of the upper bounds, or literature priority of the examples.

6.4. Other limits

The strict accessibility condition in (18) is essential. For $E_0 = E_1 = I/2$ and orthogonal pure records, $e = 1/2$, Φ is constant, $\eta = 0$, but $D_{\mathrm{rec}} = 1$. This example violates the main theorem's near-unit hypothesis; it concerns the accessibility lemma's endpoint.

The prescribed spectral selection is not globally continuous. Fix $\delta = 1/5$ and

$$E_0(\varepsilon) = \mathrm{diag}(1, 4/5 - \varepsilon, 0, 1/2), \qquad E_1(\varepsilon) = I - E_0(\varepsilon).$$

At $\varepsilon = 0$, $F_0 = \mathrm{diag}(1, 1, 0, 1/2)$. For $0 < \varepsilon < 1/10$, $F_0(\varepsilon) = E_0(\varepsilon)$. All examples satisfy the hypotheses with $\alpha = 1$. Input measurement distance 2ε tends to zero, while the completed measurement distance tends to $2/5$. A perturbation-continuity claim therefore requires additional threshold isolation.

Finally, (6) is informative only when $a + \sqrt{\delta} < 1$. The record certificate can become trivial for small κ or unfavorable η/δ. No experimental estimation procedure for all these parameters, nor practical advantage of the unweighted over the weighted completion, is established here.

7. Sharpness and reference systems

7.1. Exact envelope for orthogonal sectors

Theorem 7 (perfect packing). Under Theorem 1 with $\alpha = 1$,

$$\nu := \|\mathcal{M}_E - \mathcal{M}_F\|_\diamond \le B(\delta) := \frac{2\sqrt{\delta}}{2 - \sqrt{\delta}}. \tag{27}$$

For every $0 < \delta < 1/2$, equality is attained by a five-outcome POVM on $\mathbb{C}^6$, using a system-only input. Thus (27) is the exact worst-case envelope in this subclass, over arbitrary finite dimensions and outcome counts.

Proof. The diagonal blocks of T^*T are identities, so its trace is $\dim \mathcal{K}$. Its least eigenvalue is one; hence $T^*T = I$. The P_j are orthogonal and $Q_j = P_j$. Write $S = \sum_j P_j$, $R = I - S$, $A_j = P_j E_j P_j$, $\Delta_j = P_j - A_j$, and $\Delta = \sum_j \Delta_j$. Spectral commutation gives $B_j := E_j - A_j \ge 0$, with $\sum_j B_j = L := R + \Delta$ and $0 \le \Delta \le \delta S$. Normalize the B_j on the support of L: $N_j = L^{-1/2} B_j L^{-1/2}$. Allocate the kernel projection to any label to obtain a POVM N. Since $B_j \le L$, the factorization $B_j = \sqrt{L} N_j \sqrt{L}$ is exact, including on the kernel. Therefore

$$\mathcal{M}_E - \mathcal{M}_F = \mathcal{M}_N(\sqrt{L} \cdot \sqrt{L} - R \cdot R) - \mathcal{M}_\Delta. \tag{28}$$

Here $\mathcal{M}_\Delta$ is completely positive, though generally not trace preserving.

For any unit vector ψ, including a reference system, put $p = \|(S \otimes I)\psi\|^2$, $q = 1 - p$, and $r = \langle \psi, (\Delta \otimes I)\psi \rangle \le \delta p$. The vectors $x = (\sqrt{L} \otimes I)\psi$ and $y = (R \otimes I)\psi$ have squared norms $q + r, q$ and overlap q. Their rank-two difference has trace norm $\sqrt{r^2 + 4rq}$, by its trace r and eigenvalue product $-rq$. This standard projected-state calculation also appears explicitly in

(Regula et al. 2026, Appendix C). Contractivity and the trace r of the positive defect output give

$$\|[(\mathcal{M}_E - \mathcal{M}_F) \otimes \mathrm{id}](\psi\psi^*)\|_1 \le r + \sqrt{r^2 + 4rq} \le \delta p + \sqrt{\delta^2 p^2 + 4\delta p(1-p)}. \tag{29}$$

Set $b = \sqrt{\delta}$, $z = 2 - b$, $B = 2b/z$. Since $B - b^2 p > 0$, the identity

$$(B - b^2 p)^2 - (b^4 p^2 + 4b^2 p(1-p)) = 4b^2 (p - 1/z)^2$$

proves (27), with equality in the scalar optimization at $p = 1/z$. Reference-assisted pure inputs suffice for the diamond norm.

For attainment, put $h = \sqrt{1-b}$, so $z = 1 + h^2$, and use basis $e_0, e_1, e_2, e_3, e_4, e_r$. Let $P_j = e_j e_j^*$. On $\mathrm{span}\{e_0, e_r\}$, define

$$\begin{gathered} L_+ = \frac{1}{z}\begin{pmatrix} b^2 & bh \\ bh & h^2 \end{pmatrix}, \qquad L_- = \frac{1}{z}\begin{pmatrix} b^2 h^2 & -bh \\ -bh & 1 \end{pmatrix}, \\ E_0 = (1-b^2)P_0, \qquad E_1 = P_1 + L_+/2, \quad E_2 = P_2 + L_+/2, \\ E_3 = P_3 + L_-/2, \quad E_4 = P_4 + L_-/2. \end{gathered} \tag{30}$$

The positive rank-one operators $L_\pm$ sum to $b^2 P_0 + R \le I$. Thus the effects sum to I, and every low part has norm at most $1/2 < 1 - \delta$. Their full threshold sectors are exactly the P_j, including the closed-threshold eigenvalue of E_0. For $\psi = (e_0 + h e_r)/\sqrt{z}$, the five probability differences are $-b^2/z, b/(2z), b/(2z), -bh^2/(2z), -bh^2/(2z)$. Their absolute sum is $2b/z$, proving attainment. □

Consequently $B(\delta) = \sqrt{\delta} + \delta/2 + O(\delta^{3/2})$. The optimal asymptotic coefficient is one; the best uniform coefficient of $\sqrt{\delta}$ for $0 < \delta < 1/2$ is $(8+2\sqrt{2})/7$, approached as $\delta \uparrow 1/2$. This is sharpness for the prescribed completion, not for the best freely chosen repeatable target. If $R = 0$, (29) instead gives $\nu \le 2\delta$. The extended proof and exact attaining certificates provide additional checks independent of numerical solver tolerances.

7.2. A reference can still improve discrimination

Define $\nu_{\mathrm{sys}} = \sup_\rho \sum_j |\operatorname{Tr}((E_j - F_j)\rho)|$. Theorem 7's attaining family does not imply $\nu = \nu_{\mathrm{sys}}$ for every measurement with $\alpha = 1$. Reference-assisted measurement discrimination is an established subject (Puchała et al. 2018); the point here is its occurrence within this particular completion class.

Proposition 8 (computer-assisted reference advantage). There is a rational eight-outcome POVM on $\mathbb{C}^{10}$, satisfying Theorem 1 with $\alpha = 1$ and $\delta = 17/100$, for which

$$\nu_{\mathrm{sys}} \le 0.1161 < 0.1208 \le \nu. \tag{31}$$

Certificate proof. The specified effects and certificate supplement form part of the statement and proof. With $D_j = E_j - F_j$, $\nu_{\mathrm{sys}} = \max_{s \in \{-1,1\}^8} \lambda_{\max}(\sum_j s_j D_j)$. Exact positivity of all 256 matrices $0.1161 I - \sum_j s_j D_j^T$ gives the upper bound. Rational Hermitian SDP witnesses satisfy $\operatorname{Tr}\rho = 1$, $\rho \ge 0$, $-\rho \le X_j \le \rho$, and $\sum_j \operatorname{Tr}(D_j^T X_j) \ge 0.1208$, proving the lower bound by standard diamond-norm duality (Watrous 2012). The verifier also checks every full spectral sector and the exact prescribed target. The advantage is at least 0.0047; neither distance is asserted to be exactly equal to the displayed endpoint. □

7.3. Imperfect packing and failed extensions

Proposition 9 (coupled simplex lower bound). For every even $m \ge 4$ and $0 < \alpha \le 1$, an m-outcome POVM on $\mathbb{C}^{m+1}$ satisfies Theorem 1 with the stated α, $\delta = 1/m$, and

$$\nu \ge L_m(\alpha) := \sqrt{\frac{(m-2)[m - (m-1)\alpha^2]}{m(m-2) + m - (m-1)\alpha^2}}. \tag{32}$$

For fixed $\alpha < 1$, this lower bound tends to $\sqrt{1-\alpha^2}$ as $m \to \infty$. If $a_m^2 = 1-\alpha_m^2 = O(1/m)$, then $L_m(\alpha_m)/\sqrt{a_m^2 + 1/m} \to 1$.

Construction argument. The analytic supplement specifies the matrices and forms part of this proof. The synthesis is $cI + (\alpha - c)uu^*$, with $u = m^{-1/2}(1, \dots, 1)$ and $c = \sqrt{(m-\alpha^2)/(m-1)}$. Residual coupling vectors sum to zero and are orthogonal to their respective selected eigenvectors. Their size is fixed by a Schur complement, proving positivity. Each selected eigenvalue is $1-1/m$; its positive complementary block has trace $2/m < 1-1/m$, so the full sectors are isolated. The positive synthesis makes the polar sectors coordinate projections. Input $(u+e_r)/\sqrt{2}$ has output distance exactly $L_m(\alpha)$. The limits follow from (32). □

The exact joint finite-error envelope remains open. Two tempting extensions of (27) are already false:

$$\nu \not\leq 2(1-\alpha) + B(\delta) \quad \text{in general,} \qquad \nu \not\leq \sqrt{1-\alpha^2 + B(\delta)^2} \quad \text{in general.} \tag{33}$$

For the first, (32) with $m = 64$, $\alpha = 9/10$ has squared lower bound $80414/398097 > 1/9$, while the proposed upper bound is $1/3$. For the second, a rational seven-dimensional variant has $\alpha = 13/27$, $\delta = 1/6$, system-only distance at least 1.0405, and proposed bound at most 1.0156. The same supplement links its exact effects, rational input and sign witness, and verifier. These are counterexamples to proposed extensions, not to Theorem 1.

8. Sharp joint small-error asymptotics

The residual argument of Theorem 7 extends to nonorthogonal sectors if the selected-effect correction is estimated separately. This closes the leading joint gap without asserting either of the false finite-error extensions (33).

Theorem 10 (joint leading envelope). Under the hypotheses of Theorem 1, write

$$\begin{aligned} a^2 = 1-\alpha^2, \quad s = a^2 + \delta, \quad d = \sqrt{1-\delta}, \\ \varepsilon = 1 - \alpha^2(1-\delta), \qquad h = \max\{1-\alpha, d^{-1} - 1\} + 1 - d. \end{aligned} \tag{34}$$

Then the prescribed unweighted completion satisfies, without condition (7),

$$\nu \leq \min\left\{2, \frac{2\sqrt{\varepsilon}}{\sqrt{4-\varepsilon}} + 2h\right\}. \tag{35}$$

Let $\mathcal{W}(\alpha, \delta)$ be the supremum of ν over all finite POVMs satisfying Theorem 1 with synthesis minimum exactly α and the specified threshold δ. Dimension and outcome count may vary. For $0 < \delta \leq 1/4$ this class is nonempty for every $0 < \alpha \leq 1$, and

$$\boxed{\frac{\mathcal{W}(\alpha, \delta)}{\sqrt{1-\alpha^2+\delta}} \longrightarrow 1 \quad \text{as } (\alpha, \delta) \longrightarrow (1, 0), \quad \delta > 0.} \tag{36}$$

The limit is joint, with no fixed error ratio required. An explicit consequence is

$$(1-2\delta)\sqrt{s} \leq \mathcal{W}(\alpha, \delta) \leq \sqrt{s} + 5s \qquad (s \leq 1/4). \tag{37}$$

In particular, when $1-\alpha^2 = r\delta$ for fixed $r \geq 0$, $\mathcal{W}/\sqrt{\delta} \to \sqrt{1+r}$. The coefficient five in (37) is not asserted to be optimal.

Proof. Keep the sector operators $T = UH, D, C_j$ from Section 3. Spectral commutation and $[D, C_j] = 0$ give

$$A_j = P_j E_j P_j = UHDC_jDHU^*, \qquad B_j = E_j - A_j \geq 0.$$

As $A_j \le E_j$, their sum is $UHD^2HU^* \le I$. Congruence of $D^2 \ge d^2 I$, together with $H \ge \alpha I$, therefore yields

$$0 \le \Delta_{\mathcal{K}} := I - HD^2H \le \varepsilon I.$$

No commutation of H and D is assumed. Put $\Delta = U\Delta_{\mathcal{K}}U^*$, so $\sum_j B_j = L := R + \Delta$ and $\sqrt{L} = R + \sqrt{\Delta}$. Normalize $N_j = L^{-1/2}B_jL^{-1/2}$ on the support of L. Each B_j kills its kernel, since $0 \le B_j \le L$; allocating that kernel projection to any label extends the N_j to a POVM. Let $\mathcal{N}$ be its CPTP measurement map. Because $A_jR = 0$, $RB_jR = RE_jR$, and the exact decomposition is

$$\mathcal{M}_E - \mathcal{M}_F = \mathcal{N}(\sqrt{L} \cdot \sqrt{L} - R \cdot R) + (\mathcal{M}_A - \mathcal{M}_Q). \tag{38}$$

The maps $\mathcal{M}_A, \mathcal{M}_Q$ are CP but need not preserve trace.

Take a unit input ψ with an arbitrary reference, and set $p = \langle \psi, (S \otimes I)\psi \rangle$, $q = 1 - p$, and $t = \langle \psi, (\Delta \otimes I)\psi \rangle \le \varepsilon p$. For $x = (\sqrt{L} \otimes I)\psi$, $y = (R \otimes I)\psi$, the squared norms are $t + q, q$ and the overlap is q. Thus $\|xx^* - yy^*\|_1 = \sqrt{t^2 + 4tq}$, as in Theorem 7's established rank-two calculation (Regula et al. 2026, Appendix C). Contractivity of $\mathcal{N} \otimes \mathrm{id}$ bounds the first term of (38).

For the second term put $z = (U^* \otimes I)\psi$ and $w = (DH \otimes I)z$, so $\|z\|^2 = p$. The map DH is a contraction because $HD^2H \le I$. The two selected outputs are obtained by the same coordinate measurement of ww^* and zz^*. Moreover,

$$\|DH - I\| \le \|D(H - I)\| + \|D - I\| \le \max\{1 - \alpha, d^{-1} - 1\} + 1 - d = h.$$

The outer-product bound gives $\|ww^* - zz^*\|_1 \le (\|w\| + \|z\|)\|w - z\| \le 2hp$. Consequently every reference-assisted input obeys

$$\|[(\mathcal{M}_E - \mathcal{M}_F) \otimes \mathrm{id}](\psi\psi^*)\|_1 \le 2hp + \sqrt{\varepsilon^2 p^2 + 4\varepsilon p(1 - p)}. \tag{39}$$

For $0 < \varepsilon < 1$, the squared radical equals

$$\frac{4\varepsilon}{4 - \varepsilon} - \varepsilon(4 - \varepsilon)\left(p - \frac{2}{4 - \varepsilon}\right)^2.$$

Together with $p \le 1$, this proves (35) by the pure-input characterization of diamond distance. The selected correction is of order s, not $\sqrt{s}$, and its estimate does not sum errors over labels.

For the matching lower bound at an arbitrary $0 < \delta \le 1/4$, choose $m = 2\lceil 1/(2\delta) \rceil$. In Proposition 9's family take the requested α and $\delta' = 1/m \le \delta$. Its selected eigenvalue $1 - \delta'$ is at least $1 - \delta$; every complementary eigenvalue is at most $2/m \le 2\delta < 1 - \delta$. Thus the full sectors, synthesis minimum and completion are unchanged at the requested threshold. The class defining $\mathcal{W}$ is nonempty, and (32) gives

$$\begin{aligned} \mathcal{W}^2 \ge L_m^2 &= \frac{m - 2}{m - 1 + a^2}\left(a^2 + \frac{1}{m - 1}\right) \\ &\ge \frac{1 - 2\delta}{1 + 2\delta}(a^2 + \delta). \end{aligned} \tag{40}$$

Indeed $(m - 2)/(m - 1 + a^2) \ge 1 - 2\delta$ and $1/m \ge \delta/(1 + 2\delta)$. Meanwhile $\varepsilon = s - a^2\delta$, $\varepsilon/s \to 1$, and

$$1 - \alpha = \frac{a^2}{1 + \alpha}, \quad d^{-1} - 1 = \frac{\delta}{d(1 + d)}, \quad 1 - d = \frac{\delta}{1 + d}$$

show uniformly that $h = O(s)$. Taking square roots in (40) and dividing both bounds by $\sqrt{s}$ proves (36). For (37), when $s \le 1/4$, use $h \le 2s$, $(1 - s/4)^{-1/2} \le 1 + s$, and $\sqrt{(1 - 2\delta)/(1 + 2\delta)} \ge 1 - 2\delta$. □

Theorem 10 determines a leading worst-case envelope, not a finite-error formula for each POVM. Its lower examples use growing dimension and label count. The exact general envelope, sharp subleading term, and fixed-dimension optima remain open. Even at $\alpha = 1$, (27) exceeds $\sqrt{\delta}$ for every positive δ; the remainder in (35) cannot simply be omitted. The argument uses established tools and the specified lower construction. Its completion-specific implication and publication priority require external assessment. The extended derivation and operator checks provide supplementary detail; the upper proof and limit argument are given here.

9. Reproducibility, methodology, and review status

The RC-Discovery repository contains the frozen statement, analytic source notes, numerical identity checks, reference-assisted primal/dual SDPs, negative controls, and recorded seeds and source hashes. Its theorem-level comparison was checked on 104 weighted constructions and six primal/dual SDPs, alongside exact rational examples. The analytic proofs of Theorem 1, Corollary 3, and Propositions 4–5 do not rely on solver output. Proposition 6 is computer-assisted: rational certificates and the stated analytic error bounds establish the inequalities. Its combined audit passes 139 independent integer-determinant positivity checks and 12 rejection controls. These checks do not prove uniqueness. Theorem 7 is analytic and has seven additional exact attaining certificates. Proposition 8's audit checks 289 positive-definite matrices with 5,764 integer leading minors and 13 rejection controls. The rational veto following Proposition 9 passes 42 integer leading-minor checks and ten rejection controls. Its universal lower-bound family is established by the linked analytic construction, not extrapolation from numerical tests. A separate imperfect-packing search checked 56 structured cases, 711 accepted adaptive cases, and seven SDPs; these are numerical evidence only. Theorem 10's proof is analytic. Its separate numerical audit checks 96 operator cases, 1,440 pure inputs with reference dimensions 1, 2 and the system dimension, twenty structured witnesses, and five full diamond SDPs. These checks probe the decomposition and estimates; they do not prove the universal limit or replace its arbitrary-reference argument. The Lean development checks 13 constituent declarations, not the full spectral and channel theorem.

This work was developed using **VIBE MATH**, an AI-assisted workflow of conjecture generation, derivation, adversarial checking, literature comparison, and computational support directed by Christopher Gardner. AI systems are research tools, not authors or endorsers. Agreement among internal AI-assisted checks is not external verification.

The requested external assessment is concrete: check the supplied reduction of Theorem 1 and Corollary 3 to established tools, reconstruct the sharp subclass theorem, joint asymptotics, lower-bound family and exact comparisons, and identify any earlier statement or application that subsumes this quantitative specialization. The reduction requires no new general lemma; publication priority and usefulness remain unverified. Correctness, priority, optimality, and usefulness are separate questions. This draft has not undergone independent specialist review and is not represented as submission-ready.

References

Bruijn, N. G. de. 1980. "Remarks on Hermitian Matrices." *Linear Algebra and Its Applications* 32: 201–8. https://doi.org/10.1016/0024-3795(80)90017-8.

Busch, Paul, Marian Grabowski, and Pekka J. Lahti. 1995. "Repeatable Measurements in Quantum Theory: Their Role and Feasibility." *Foundations of Physics* 25 (9): 1239–66. https://doi.org/10.1007/BF02055331.

Eldar, Yonina C., and G. David Forney Jr. 2001. "On Quantum Detection and the Square-Root Measurement." *IEEE Transactions on Information Theory* 47 (3): 858–72. https://doi.org/10.1109/18.915636.

Higham, Nicholas J. 1986. "Computing the Polar Decomposition—with Applications." *SIAM Journal on Scientific and Statistical Computing* 7 (4): 1160–74. https://doi.org/10.1137/0907079.

Kitaev, Alexei. 2025. *Almost-Idempotent Quantum Channels and Approximate C^*-Algebras*. arXiv:2405.02434v2. https://doi.org/10.48550/arXiv.2405.02434.

Kotowski, Michał, and Michał Oszmaniec. 2025. *Pretty-Good Simulation of All Quantum Measurements by Projective Measurements*. arXiv:2501.09339v1. https://doi.org/10.48550/arXiv.2501.09339.

Kretschmann, Dennis, Dirk Schlingemann, and Reinhard F. Werner. 2008. "A Continuity Theorem for Stinespring's Dilation." *Journal of Functional Analysis* 255 (8): 1889–904. https://doi.org/10.1016/j.jfa.2008.07.023.

Puchała, Zbigniew, Łukasz Pawela, Aleksandra Krawiec, and Ryszard Kukulski. 2018. "Strategies for Optimal Single-Shot Discrimination of Quantum Measurements." *Physical Review A* 98: 042103. https://doi.org/10.1103/PhysRevA.98.042103.

Regula, Bartosz, Ludovico Lami, and Nilanjana Datta. 2026. *Tight Relations and Equivalences between Smooth Relative Entropies*. arXiv:2501.12447v5. https://arxiv.org/abs/2501.12447v5.

Salle, Mikael de la. 2022. "Orthogonalization of Positive Operator Valued Measures." *Comptes Rendus. Mathématique* 360: 549–60. https://doi.org/10.5802/crmath.326.

Watrous, John. 2012. *Simpler Semidefinite Programs for Completely Bounded Norms*. arXiv:1207.5726v2. https://arxiv.org/abs/1207.5726v2.

Watrous, John. 2018. *The Theory of Quantum Information*. Cambridge University Press. https://doi.org/10.1017/9781316848142.